

ID: enc-01

Question: Which technologies does the Google Cloud encryption document name for protecting data in transit?

Ground Truth: It names Transport Layer Security (TLS), BoringSSL, Application Layer Transport Security (ALTS), and the PSP security protocol. It also lists IPsec, managed TLS certificates, and Cloud Service Mesh as additional protection options.

Source Document: encryption_google_cloud.pdf

Source Pages: 1

Source Section: Introduction

Category: terminology_hybrid

Difficulty: easy

ID: enc-02

Question: How does encryption in transit protect a communication if it is intercepted?

Ground Truth: The document says the endpoints are authenticated and the data is encrypted before transmission. On arrival, the receiver decrypts the data and verifies that it was not modified during transit.

Source Document: encryption_google_cloud.pdf

Source Pages: 1

Source Section: Authentication, integrity, and encryption

Category: semantic_retrieval

Difficulty: medium

ID: enc-03

Question: What functions does Google Front End (GFE) perform for incoming connections?

Ground Truth: GFE terminates incoming HTTP(S), TCP, and TLS connections, provides DDoS countermeasures, and routes and load-balances traffic to Google Cloud services.

Source Document: encryption_google_cloud.pdf

Source Pages: 3

Source Section: End user to a Google Cloud service

Category: exact_fact

Difficulty: easy

ID: enc-04

Question: Which TLS implementation does GFE use, and what FIPS validation does the document report for BoringCrypto?

Ground Truth: GFE implements TLS with Google's open-source BoringSSL. The document states that BoringCrypto, BoringSSL's core, is validated to FIPS 140-3 Level 1.

Source Document: encryption_google_cloud.pdf

Source Pages: 3

Source Section: End user to a Google Cloud service

Category: terminology_hybrid

Difficulty: medium

ID: enc-05

Question: When does the document say a Layer 7 load balancer encrypts the user connection by default, and how is TLS terminated?

Ground Truth: When the end user's client supports TLS, the connection to an external Layer 7 load balancer is encrypted with TLS by default. GFE terminates the user's TLS connection using a self-managed or Google-managed certificate.

Source Document: encryption_google_cloud.pdf

Source Pages: 4

Source Section: End user to a customer application hosted on Google Cloud

Category: exact_fact

Difficulty: medium

ID: enc-06

Question: How are private-IP VM connections within a VPC protected, and what caveat applies to VM connections using external IP addresses?

Ground Truth: Private-IP traffic within the same VPC or across peered VPC networks is encrypted at the network layer. Communicating hosts establish a session key over an ALTS-protected control channel, and the keys are periodically rotated. Connections using external IP addresses are not automatically encrypted.

Source Document: encryption_google_cloud.pdf

Source Pages: 5

Source Section: Google Cloud virtual network authentication and encryption

Category: multi_hop

Difficulty: hard

ID: enc-07

Question: What does PSP provide when SmartNIC hardware is available, and which non-TCP example and key property are specified?

Ground Truth: When SmartNICs are available, ALTS uses PSP to encrypt data in transit across Google's network and can offload encryption to SmartNIC hardware. PSP supports non-TCP protocols such as UDP and uses a unique encryption key for each connection.

Source Document: encryption_google_cloud.pdf

Source Pages: 7

Source Section: Network encryption using PSP

Category: terminology_hybrid

Difficulty: hard

ID: enc-08

Question: Which key-exchange and encryption details does the document specify for ALTS?

Ground Truth: The ALTS handshake combines elliptic-curve and quantum-safe key exchange, and supported shared-key derivation algorithms include ML-KEM (FIPS 203). The record protocol uses the shared traffic key; ALTS defaults to AES-128-GCM, while data in Google's lowest-level storage system uses AES-256.

Source Document: encryption_google_cloud.pdf

Source Pages: 8

Source Section: Application layer transport security

Category: terminology_hybrid

Difficulty: hard

ID: sec-01

Question: What responsibilities does Google's dedicated security team have, according to the overview?

Ground Truth: The team maintains defense systems, develops security review processes, builds security infrastructure, implements security policies, scans for vulnerabilities, and conducts penetration tests, quality assurance, and security reviews.

Source Document: google_cloud_security.pdf

Source Pages: 4

Source Section: Our dedicated security team

Category: exact_fact

Difficulty: medium

ID: sec-02

Question: What amount did Google's Vulnerability Reward Program award researchers in 2023?

Ground Truth: Google says it awarded researchers over \$10 million in prize money in 2023 through the Vulnerability Reward Program.

Source Document: google_cloud_security.pdf

Source Pages: 5

Source Section: Collaboration with the security research community

Category: exact_fact

Difficulty: easy

ID: sec-03

Question: How does Google's vulnerability management process check whether vulnerability fixes have actually reached systems?

Ground Truth: Automation pipelines continually reassess patch deployment to mitigate vulnerabilities and flag deployments that are incorrect or partial.

Source Document: google_cloud_security.pdf

Source Pages: 7

Source Section: Vulnerability management

Category: semantic_retrieval

Difficulty: medium

ID: sec-04

Question: Which information sources are brought together for Google's unified security monitoring, and what examples of suspicious activity are checked?

Ground Truth: The program aggregates security telemetry in one location, including internal network traffic, employee actions on systems, and outside vulnerability information. It inspects network traffic for signs such as botnet connections and examines system logs for unusual behavior such as attempts to access customer data.

Source Document: google_cloud_security.pdf

Source Pages: 8

Source Section: Security monitoring

Category: multi_hop

Difficulty: hard

ID: sec-05

Question: What incident-response guidance does Google's security incident program align with, and when is the security team available to employees?

Ground Truth: The incident-management program aligns with NIST SP 800-61 guidance. The Google security team is available to employees 24/7 to help resolve incidents.

Source Document: google_cloud_security.pdf

Source Pages: 8

Source Section: Incident management

Category: terminology_hybrid

Difficulty: easy

ID: sec-06

Question: What controls restrict access to the data-center floor?

Ground Truth: Access to the data-center floor is through a security corridor with multi-factor access control using security badges and biometrics; only approved employees with specific roles may enter.

Source Document: google_cloud_security.pdf

Source Pages: 9

Source Section: State-of-the-art data centers

Category: exact_fact

Difficulty: easy

ID: sec-07

Question: At which layers does Google Cloud apply its default encryption for data at rest?

Ground Truth: The overview says Google Cloud applies several layers of encryption by default to stored user data. Encryption is applied at the application layer, the storage-device layer, or both.

Source Document: google_cloud_security.pdf

Source Pages: 11

Source Section: Securing data at rest

Category: exact_fact

Difficulty: easy

ID: sec-08

Question: How does Confidential Computing protect data while it is being processed, and what does its attestation establish?

Ground Truth: Confidential Computing uses hardware isolation and a Trusted Execution Environment (TEE) to perform computation in cryptographic isolation. The TEE provides independently verifiable attestations of the system state and the code that ran.

Source Document: google_cloud_security.pdf

Source Pages: 12

Source Section: Securing data in use

Category: terminology_hybrid

Difficulty: medium

ID: ms-01

Question: What three core business priorities does Microsoft name in the annual report letter?

Ground Truth: Microsoft names security, quality, and AI innovation as its three core business priorities.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 1

Source Section: Our Priorities

Category: exact_fact

Difficulty: easy

ID: ms-02

Question: How many full-time-equivalent engineers did Microsoft dedicate through SFI to its highest-priority security work?

Ground Truth: Microsoft says it dedicated the equivalent of 34,000 full-time engineers to its highest-priority security work through the Secure Future Initiative (SFI). The work included strengthening identity protection, securing networks and systems, improving threat detection and response, and embedding secure-by-design practices.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 1

Source Section: Our Priorities — Security

Category: terminology_hybrid

Difficulty: medium

ID: ms-03

Question: How many datacenters and regions did Microsoft report operating, and how much new capacity did it add that year?

Ground Truth: Microsoft reported operating more than 400 datacenters in 70 regions and adding over two gigawatts of new capacity during the year.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 2

Source Section: Our Cloud and AI infrastructure

Category: exact_fact

Difficulty: easy

ID: ms-04

Question: What makes OneLake a foundation for enterprise AI applications in Microsoft's description?

Ground Truth: The report says OneLake spans databases and clouds, including Power BI semantic models, and describes this as a foundation for building enterprise AI applications. It also says Microsoft Fabric had 25,000 paid customers.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 2

Source Section: Our Cloud and AI infrastructure

Category: semantic_retrieval

Difficulty: medium

ID: ms-05

Question: What did Azure AI Foundry provide at introduction, and what adoption figure did Microsoft report for Fortune 500 companies?

Ground Truth: Azure AI Foundry was described as a platform to design, customize, and run AI applications and agents, with access to more than 11,000 partner models. Microsoft said 80 percent of the Fortune 500 used Foundry for AI workloads.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 2

Source Section: Our Cloud and AI infrastructure

Category: terminology_hybrid

Difficulty: medium

ID: ms-06

Question: How many monthly active users did Microsoft report across its commercial and consumer Copilot products?

Ground Truth: Microsoft reported more than 100 million monthly active users across its commercial and consumer Copilot products.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 2

Source Section: Copilots and agents

Category: exact_fact

Difficulty: easy

ID: ms-07

Question: What adoption figures does the report give for Copilot Studio and GitHub Copilot?

Ground Truth: More than 230,000 organizations used Copilot Studio to extend Microsoft 365 Copilot or build agents, and GitHub Copilot had more than 20 million users.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 3

Source Section: Copilots and agents

Category: exact_fact

Difficulty: easy

ID: ms-08

Question: How does Microsoft 365 Commercial combine products for organizations, and what factors does the report say drive its revenue growth?

Ground Truth: Microsoft 365 Commercial brings together Office, Windows, Microsoft 365 Copilot, and Enterprise Mobility + Security. Growth depends on reaching new users, adding AI-enabled tools and agentic scenarios, expanding the installed base and average revenue per user, and the continued shift from on-premises Office licenses to Microsoft 365.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 12

Source Section: Microsoft 365 Commercial Products and Cloud Services

Category: multi_hop

Difficulty: hard

ID: ms-09

Question: What were Microsoft's fiscal 2025 revenue and operating income, and how did each change from fiscal 2024?

Ground Truth: Microsoft reported fiscal 2025 revenue of \$281.724 billion, up 15 percent from \$245.122 billion in fiscal 2024. Operating income was \$128.528 billion, up 17 percent from \$109.433 billion.

Source Document: microsoft_annual_report_2025.pdf

Source Pages: 23

Source Section: Summary Results of Operations

Category: exact_fact

Difficulty: medium

ID: cross-01

Question: How does the Google Cloud Security Overview's description of user-to-GFE protection compare with the more detailed Encryption in Transit document's account of traffic to and from services?

Ground Truth: The overview says traffic between user devices and Google Front End (GFE) is encrypted with strong protocols such as TLS. The detailed document says user traffic to GFE is encrypted with TLS or QUIC, while ALTS provides authenticated, integrity-protected encryption between GFE and Google Cloud services and between services.

Source Document: google_cloud_security.pdf; encryption_google_cloud.pdf

Source Pages: 11, 6

Source Section: Securing data in transit; Connectivity to Google APIs and services; Service-to-service authentication, integrity, and encryption

Category: cross_document

Difficulty: hard

ID: cross-02

Question: How does Google's shared-responsibility description differ in focus from Microsoft's Secure Future Initiative work described in its annual report?

Ground Truth: Google's overview assigns customers responsibility for securing what they bring to the cloud, including their data, while Google secures the underlying cloud infrastructure; it also distinguishes IaaS and SaaS responsibilities. Microsoft's report describes internal SFI work, including a 34,000-engineer equivalent focused on identity protection, networks and systems, threat detection and response, and secure-by-design practices.

Source Document: google_cloud_security.pdf; microsoft_annual_report_2025.pdf

Source Pages: 17, 1

Source Section: Google Cloud security products and services; Our Priorities — Security

Category: cross_document

Difficulty: hard

ID: neg-01

Question: What total revenue did Google report for fiscal year 2025?

Ground Truth: The supplied documents do not state Google's fiscal year 2025 total revenue.

Source Document: All supplied PDFs

Source Pages: Not stated

Source Section: Not stated in the supplied corpus

Category: unsupported

Difficulty: medium

ID: neg-02

Question: How many Google Cloud datacenters does Google operate worldwide?

Ground Truth: The supplied documents do not give a worldwide count of Google Cloud datacenters.

Source Document: All supplied PDFs

Source Pages: Not stated

Source Section: Not stated in the supplied corpus

Category: unsupported

Difficulty: medium

ID: neg-03

Question: What was Microsoft's exact Azure revenue in fiscal year 2024?

Ground Truth: The supplied documents do not state an exact fiscal year 2024 Azure revenue amount.

Source Document: All supplied PDFs

Source Pages: Not stated

Source Section: Not stated in the supplied corpus

Category: unsupported

Difficulty: medium
